

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

May 2014

CL 404 Water Resources Engineering & Management

Date: 23.05.2014, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures with pencil only wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q – 1(a) Explain the importance of the rivers in the development human civilization. Also, [04]
distinguish between perennial river and virgin river.

(b) Explain the process of 'River meandering'. [03]

Q – 2(a) Discuss the importance of capacity-elevation and area-elevation curves of a reservoir [03]
site in the reservoir planning.

(b) Following data has been collected during contour survey of a reservoir site: [07]

R.L. (m)	Area (ha.)	Capacity (ha. m)
110.00	10	21.20
125.00	21	?
140.00	36	?
155.00	?	?

Develop the equations for Area-elevation curve and Capacity-elevation curve for given data. Also calculate the missing data (values) of the table mentioned above,

(c) What is river training works? What are the objectives of river training works? [04]

OR

Q – 2(a) What is a mass curve of inflow? How is it used to determine the storage capacity of a [06]
reservoir for a constant demand?

(b) Discuss, with a suitable sketch, different zones of storage of the reservoir. [04]

(c) List the types of river training works and discuss in detail ONE of them. [04]

- Q – 3(a) Differentiate, clearly, between Thermal power plant and Hydropower plant. [04]
- (b) Explain the following terms: [06]
- (i) Load factor (ii) Plant factor (iii) Firm power
- (c) What is drought? Suggest different measures for drought protection. [04]

OR

- Q – 3(a) Explain: Surge tank. What is its importance in a hydroelectric project? [03]
- (b) Draw a neat sketch showing different components of a hydro electric project. Discuss the functions of each of them. [07]
- (c) What are the structural methods of flood control? Discuss ONE of them with a neat sketch. [04]

SECTION – II

- Q – 4 (a) What is reservoir sedimentation? Also, explain 'density current' in view of reservoir sedimentation. [03]
- (b) Explain the different types of investigation necessary for the planning of a reservoir? [04]
- Q - 5 Attempt any Two: [14]
- (a) The yearly data for a catchment of a proposed reservoir site for 30 years is given in following table. Compute the dependable rainfall for 55% & 70% dependability.

Year	1945	1946	1947	1948	1949	1950	1951	1952	1953	1954
Rainfall (cm)	78	76	101	112	80	85	116	120	140	150
Year	1955	1956	1957	1958	1959	1960	1961	1962	1963	1964
Rainfall (cm)	99	98	66	70	104	92	95	87	69	91
Year	1965	1966	1967	1968	1969	1970	1971	1972	1973	1974
Rainfall (cm)	131	138	126	128	120	111	110	83	82	94

- (b) Discuss planning strategy for water resources development projects: List out various steps involved to plan WRP. Explain in detail data required to be collected for planning of a WRP.
- (c) Which measures have been taken to control the sedimentation before and after constructing the reservoir?

Q - 6 Attempt any Two:

[14]

- (a) For better planning and management of project which strategies are carried out?
- (b) The runoff from a catchment area during successive months in a year is given below. Determine the maximum capacity of the reservoir required if the entire volume of water is to be drawn off at a uniform rate, without any loss of water over the spillway.

Use Analytical method only.

Month	Jan	Feb	Mar	April	May	June	July	Aug	Sept	Oct	Nov	Dec
Runoff (Mm ³)	1.3	2.0	2.70	8.50	12.0	12.0	19.0	22.0	2.50	2.20	1.90	1.70

- (c) Which curves are generally plotted for the economic analysis of WRP? Draw each of them and explain how the decision about the size of project can be made?
